

Week 2, September 11, 2024

Main topics

- Numerical implementation of model problems
- Singular value decomposition of operators and matrices
- Truncated SVD
- Regularization strategy

Material

From last week: Section 1.1-1.2 in [Kirsch] and section 3.1 in [PCH].

New material: Kirsch 2-2.1 (you may skip 27-35). Appendix A, in particular Definition A.56, Theorem A.57, Theorem A.58. You can think of compact operators as integral operators.

Program

We will go with the tentative program:

13:00-13:30 Ill-posedness and model problems

13:30-15:30 Exercise session - discretization of model problems

15:45-17:00 SVD for matrices and compact operators; regularization

Exercises

You are highly encouraged to discuss and solve the exercises in small groups.

In the exercise session today we will implement the model problems and approximate the forward operator K by a matrix A .

1. Warm up: (spend less than 15 minutes)

The numerical differentiation problem is an inverse problem (see slides from the morning in week 1). The problem is implemented in the code `NumericalDifferentiation.m` to be found in learn (see the module Matlab). Try out the code.

- Change the function to $y(s) = s^2 e^{-s}$.
- Solve the inverse problem and compare to the true solution.
- Experiment with the noise level and the number of discretization points.

2. Abel operator by quadrature and collocation (Nyström method):

The operator

$$Kx(t) = \frac{1}{\sqrt{\pi}} \int_0^t \frac{x(s)}{\sqrt{t-s}} ds$$

$$= \int_0^1 k(t, s)x(s) ds \quad \text{with } k(t, s) = \begin{cases} \frac{1}{\sqrt{\pi(t-s)}}, & s < t \\ 0, & s > t \end{cases}$$

is approximated by a matrix A as follows: Take $h = 1/N$ and define

s -grid: $s_1 = h/2, s_2 = h/2 + h, \dots, s_N = h/2 + (N-1)h$

t -grid $t_1 = h, t_2 = 2h, \dots, t_N = Nh = 1$.

Then set $A_{ij} = hk(t_i, s_j)$, i.e. apply

$$Kf(t_i) \approx h \sum_j k(t_i, s_j)f(s_j).$$

- Implement the method numerically to get the matrix A . Experiment with different functions x .
- Logplot the singular values of the A . (Use `[U, S, V] = svd(A)` in matlab.)
- Try having similar t and s grids. Find a way for handling the singularity.

3. Heat equation by discrete Fourier transform

We will use the variables (s, t) ; s is the spatial variable for initial condition $x(s)$, t is the spatial variable for the terminal condition $y(t)$. Recall the formula

$$Kx(t) = y(t) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} c_n e^{-Tn^2} \sin(nt),$$

with

$$x(s) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} c_n \sin(ns)$$

being the initial condition with Fourier (sine) coefficients c_n .

Put differently the operator K is

$$y(t) = Kx(t) = \int_0^\pi k(t, s)x(s) ds$$

with

$$k(t, s) = \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-Tn^2} \sin(nt) \sin(ns).$$

K is implemented by the discrete sine transform via FFT, e.g. in Matlab by the following snippets

```

% Fourier modes
k = [-N:N-1]; % Wave numbers
k = fftshift(k); %Shift frequencies to match FFT

% odd extension to force full Fourier series gives sine
series
u = u0(2:end-1);
u_ext = [0,u,0,-flip(u)];

% Compute Fourier coeff
u_hat = fft(u_ext); % FFT of the solution
u_hatT = u_hat .* exp(-k.^2 * T); % Multiply each
    Fourier coefficient by the right exponential

uend = ifft(u_hatT); % Inverse FFT to get back to
    physical space

```

- Set up a complete code that given initial function x approximates y (both are sampled and represented as vectors)
- Experiment with different choices of initial conditions x and compute an approximation of y . Look at different choices of T .
- The problem is linear and can be represented by a matrix A . Compute the matrix.
- Logplot the singular values of A .
- How would you characterize the method as a quadrature + collocation or expansion method.

4. Deblurring problem by Galerkin method:

Define

$$k(z) = \begin{cases} 1 + \cos(\pi z/3), & |z| < 3, \\ 0, & |z| \geq 3. \end{cases}$$

Approximate numerically the convolution operator K in $L^2(-6, 6)$

$$Kx(t) = \int_{-6}^6 k(t-s)x(s)ds \quad (1)$$

by a Galerkin scheme as follows:

- Take n with $n/4$ integer and put $h = 12/n$. Define the z -grid by $z_j = -6 + jh$, $j = 0, \dots, n$.
- Take for $j = 1, \dots, n$ basis functions $\phi_j(z) = 1/\sqrt{h}$ for $z \in (z_{j-1}, z_j)$ and 0 otherwise and expand both x and $y = Kx$ in that basis.

- Calculate the matrix elements

$$A_{ij} = \int_{-6}^6 \int_{-6}^6 \phi_i(t) k(t-s) \phi_j(s) ds dt.$$

Hint: Start by fixing $j - 1 = n/2$ leaving $z_{j-1} = 0$.

- The matrix A now takes the expansion coefficients of x to expansion coefficients of y . Interpret the matrix in the particular basis as a mapping from x sampled on the z -grid to the corresponding y .